

# 1. Rational and Irrational numbers

① Natural Numbers :- 'N' = 1, 2, 3, 4, ...

② Whole Numbers :- 'W' = 0, 1, 2, 3, ...

③ Prime Numbers :- 'P' = 2, 3, 5, 7, ...

④ Even Numbers :- 'E' = 0, 2, 4, 6, 8, ...

⑤ Odd Numbers :- 'O' = 1, 3, 5, 7, ...

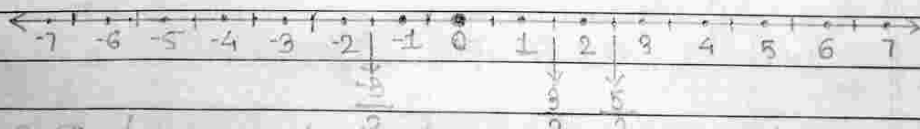
⑥ Integers :- 'I' = -3, -2, -1, 0, 1, 2, 3

Rational Numbers :- The numbers  $\frac{p}{q}$  where  $q \neq 0$  is called as rational numbers.

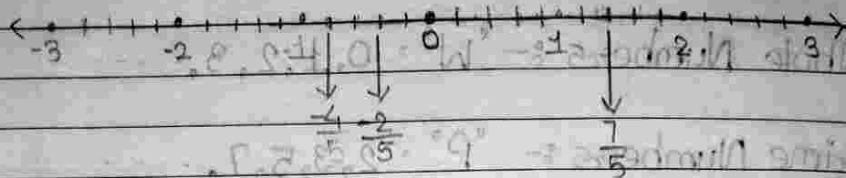
## Practice Set 1.1

1. Show the following numbers on a number line. Draw a separate number line for each example.

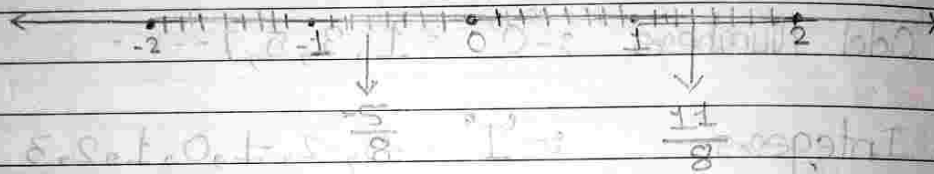
(1)  $\frac{3}{2}, \frac{5}{2}, -\frac{3}{2}$



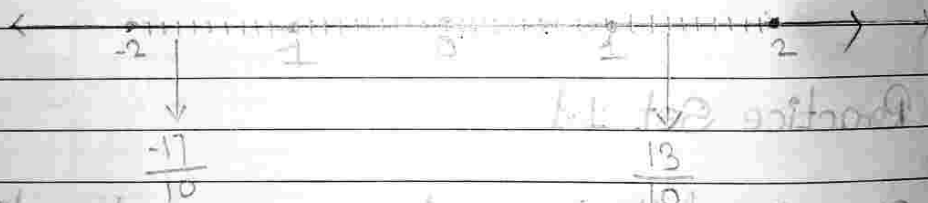
(2)  $\frac{7}{5}, -\frac{2}{5}, -\frac{4}{5}$



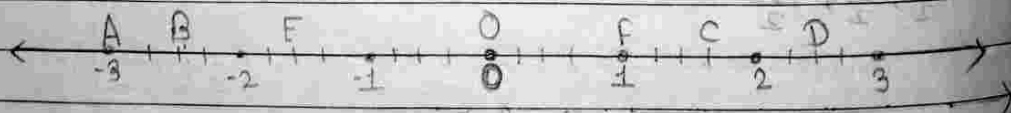
(3)  $-\frac{5}{8}, \frac{11}{8}$



(4)  $\frac{13}{10}, -\frac{17}{10}$



2. Observe the numbers line and answer the questions.



- 1) Which numbers is indicated by point B?  
 ⇒ Point B indicated numbers  $-\frac{5}{2}$

2) Which point indicated the number  $1\frac{3}{4}$ ? OP (a)  
 $\Rightarrow$  Point C indicated number  $1\frac{3}{4}$  PE PE

3) State whether the statement, 'the point D denotes the number  $\frac{5}{2}$ ', is true or false.  
 $\Rightarrow$  True.

### Comparison of rational numbers

If  $\frac{a}{b}$  and  $\frac{c}{d}$  are rational numbers

Then,

①  $ad > bc \therefore \frac{a}{b} > \frac{c}{d}$

②  $ad < bc \therefore \frac{a}{b} < \frac{c}{d}$

③  $ad = bc \therefore \frac{a}{b} = \frac{c}{d}$

### Practice Set 1.2

1. Compare the following numbers.

|                                       |                                                         |                                                       |                                                                             |
|---------------------------------------|---------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|
| (1) $-7, -2$<br>$\Rightarrow -7 < -2$ | (2) $0, -\frac{9}{5}$<br>$\Rightarrow 0 > -\frac{9}{5}$ | (3) $\frac{8}{7}, 0$<br>$\Rightarrow \frac{8}{7} > 0$ | (4) $-\frac{5}{4}, \frac{1}{4}$<br>$\Rightarrow -\frac{5}{4} < \frac{1}{4}$ |
|---------------------------------------|---------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|

$$(5) \frac{40}{29}, \frac{141}{29}$$

$$\Rightarrow \frac{40}{29} < \frac{141}{29}$$

$$(6) \frac{-17}{20}, \frac{-13}{20}$$

$$\Rightarrow \frac{-17}{20} < \frac{-13}{20}$$

$$(7) \frac{15}{12}, \frac{7}{16}$$

$$\Rightarrow 15 \times 16 > 12 \times 7$$

$$\frac{15}{12} > \frac{7}{16}$$

$$(8) \frac{-25}{8}, \frac{-9}{4}$$

$$\Rightarrow -25 \times 4 < 8 \times -9$$

$$-100 < -72$$

$$\frac{-25}{8} < \frac{-9}{4}$$

$$(9) \frac{12}{15}, \frac{3}{5}$$

$$\Rightarrow 12 \times 5 > 15 \times 3$$

$$60 > 45$$

$$\frac{12}{15} > \frac{3}{5}$$

$$(10) \frac{-7}{11}, \frac{-3}{4}$$

$$\Rightarrow -7 \times 4 < 11 \times -3$$

$$-28 < -33$$

$$\frac{-7}{11} > \frac{-3}{4}$$

Decimal representation of rational numbers

Practice Set 1.3

1. Write the following rational number in decimal form.

$$(1) \frac{9}{37} =$$

$$\Rightarrow 0.243$$

$$37 \overline{) 9.000}$$

$$\underline{-0}$$

$$90$$

$$\underline{-74}$$

$$160$$

$$\underline{-148}$$

$$0120$$

$$\underline{-111}$$

$$0090$$

$$(2) \frac{18}{42} =$$

$$42 \overline{) 18.0000}$$

$$\underline{-0}$$

$$180$$

$$\underline{-168}$$

$$0120$$

$$\underline{-84}$$

$$0460$$

$$\underline{-336}$$

$$124$$

$$\underline{-84}$$

$$40$$

$$(2) \frac{18}{42} = \frac{3}{7} \quad 7 \overline{) 3}$$

$$\Rightarrow \frac{18}{42} = 0.42871 \quad \begin{array}{r} -0 \\ 30 \\ -28 \\ \hline 020 \end{array}$$

$$\begin{array}{r} -14 \\ \hline 060 \end{array}$$

$$\begin{array}{r} -56 \\ \hline 040 \end{array}$$

$$\begin{array}{r} -35 \\ \hline 050 \end{array}$$

$$\begin{array}{r} -49 \\ \hline 010 \end{array}$$

$$\begin{array}{r} -7 \\ \hline 30 \end{array}$$

$$(3) \frac{9}{14} = \frac{0.42871}{14 \overline{) 9}}$$

$$\Rightarrow \frac{9}{14} = 0.42871 \quad \begin{array}{r} -0 \\ 90 \\ -84 \\ \hline 60 \end{array}$$

$$\begin{array}{r} -56 \\ \hline 040 \end{array}$$

$$\begin{array}{r} -28 \\ \hline 120 \end{array}$$

$$\begin{array}{r} -112 \\ \hline 0080 \end{array}$$

$$\begin{array}{r} -70 \\ \hline 100 \end{array}$$

$$\begin{array}{r} -98 \\ \hline 020 \end{array}$$

$$\begin{array}{r} -14 \\ \hline 060 \end{array}$$

$$(4) \frac{-103}{5} = -20.6 \quad \begin{array}{r} 20.6 \\ 5 \overline{) 103} \\ -10 \\ \hline 003 \end{array}$$

$$\Rightarrow \frac{-103}{5} = -20.6 \quad \begin{array}{r} -0 \\ 30 \\ -30 \\ \hline 00 \end{array}$$

$$(5) \frac{-11}{13} = -0.846153 \quad \begin{array}{r} 0.846153 \\ 13 \overline{) 11} \end{array}$$

$$\frac{-11}{13} = -0.846153 \quad \begin{array}{r} -0 \\ 110 \\ -104 \\ \hline 0060 \\ -52 \\ \hline 80 \\ -78 \\ \hline 20 \\ -13 \\ \hline 70 \\ -65 \\ \hline 050 \\ -39 \\ \hline 110 \end{array}$$